Research Methodology:

An Introduction:

MEANING OF RESEARCH: Research in common parlance refers to a search for knowledge.Once can also define research as a scientific & systematic search for pertinent information on a specific topic.In fact, research is an art of scientific investigation.

Research is an academic activity and as such the term should be used in a technical sense.

According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organizing and evaluating data; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis.

OBJECTIVES OF RESEARCH

1. To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as exploratory or formulative research studies); 2. To portray accurately the characteristics of a particular individual, situation or a group(studies with this object in view are known as descriptive research studies); 3. To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies); 4. To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies).

TYPES OF RESEARCH (i) Descriptive vs. Analytical:

Descriptive research includes surveys and fact-finding enquiries of different kinds.In social science and business research we quite often use the term Ex post facto research for descriptive research studies.The main characteristic of this method is that the researcher has no control over the variables; he can only report what has happened or what is happening.The methods of research utilized in descriptive research are survey methods of all kinds, including comparative and correlation methods.

In analytical research, on the other hand, the researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material.

(ii) Applied vs. Fundamental:

Applied (or action) research : aims at finding a solution for an immediate problem facing a society or an industrial/business organization,Fundamental (to basic or pure) research: is mainly concerned with generalizations and with the formulation of a theory.

(iii) Quantitative vs. Qualitative:
Quantitative research is based on the measurement of quantity or amount.Qualitative research, on the other hand, is concerned with qualitative phenomenon, i.e., phenomena relating to or involving quality or kind.

(iv) Conceptual vs. Empirical:
Conceptual research :is that related to some abstract idea(s) or theory.Empirical research ;It is data-based research, coming up with conclusions which are capable of being verified by observation or experiment. We can also call it as experimental type of research.

(v) Some Other Types of Research:
One-time research or longitudinal research. In the former case the research is confined to a single time-period, whereas in the latter case the research is carried on over several time-periods.Field-setting research or laboratory research or simulation research, depending upon the environment in which it is to be carried out.Clinical or diagnostic research. Such research follow case-study methods or in-depth approaches to reach the basic causal relations.Conclusion-oriented and decision-oriented. While doing conclusion-oriented research, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes.

The exploratory research or it may be formalized.
The objective of exploratory research is the development of hypotheses rather than their testing, whereas formalized research studies are those with substantial structure and with specific hypotheses to be tested.Historical research is that which utilizes historical sources like documents, remains, etc. to study events or ideas of the past, including the philosophy of persons and groups at any remote point of time.

Research ApproachesTwo basic approaches to research, viz., quantitative approach and the qualitative approach .Quantitative approach sub-classified into inferential , experimental and simulation approaches to research.Qualitative approach to research is concerned with subjective assessment of attitudes, opinionsand behavior.

Research Methods versus Methodology
Research methods or techniques, thus, refer to the methods the researchers use in performing research operations.Thus, when we talk of research methodology we not only talk of the research methods but also consider the logic behind the methods we use in the context of our research study and explain why we are using a particular method or technique and why we are not using others so that research results are capable of being evaluated either by the researcher himself or by others.

the difference between methods and techniques of data collection

research process guideline:
1formulating the research problem;(2) extensive literature survey;(3) developing the hypothesis;(4) preparing the research design;(5) determining sample design;(6) collecting the data;(7) execution of the project;(8) analysis of data;(9) hypothesis testing;(10) generalisations and interpretation,(11) preparation of the report or presentation of the results ,i. e., formal write-up of conclusions reached.

1. Formulating the research problem:
There are two types of research problems, viz., those which relate to states of nature and those which relate to relationships between variables.

2. Extensive literature survey:
Once the problem is formulated, a brief summary of it should be written down. It is compulsory for a research worker writing a thesis for a Ph.D. degree to write a synopsis of the topic and submit it to the necessary Committee or the Research Board for approval.

3. Development of working hypotheses:
After extensive literature survey, researcher should state in clear terms the working hypothesis or hypotheses.Working hypothesis is tentative assumption made in order to draw out and test its logical or empirical consequences.

4. Preparing the research design:
The research problem having been formulated in clear cut terms, the researcher will be required to prepare a research design, i.e., he will have to state the conceptual structure within which research would be conducted.Research purposes may be grouped into four categories, viz., (i) Exploration, (ii) Description, (iii) Diagnosis, and (iv) Experimentation.

There are several research designs, such as, experimental and non-experimental hypothesis testing.
Experimental designs can be eitherInformal designs (such as before-and-after without control, after-only with control, before-and-after with control)Formal designs (such as completely randomized design, randomized block design, Latin square design, simple and complex factorial designs), out of which the researcher must select one for his own project.

5. Determining sample design:
All the items under consideration in any field of inquiry constitute a ‘universe’ or

'population'.A complete enumeration of all the items in the 'population' is known as a census inquiry.A brief mention of the important sample designs is as follows:(i) Deliberate sampling: Deliberate sampling is also known as purposive or non-probability sampling.

(ii) Simple random sampling:
This type of sampling is also known as chance sampling or probability sampling where each and every item in the population has an equal chance of inclusion in the sample and each one of the possible samples, in case of finite universe, has the same probability of being selected.

(iii) Systematic sampling: In some instances the most practical way of sampling is to select every 15th name on a list, every 10th house on one side of a street and so on.(iv) Stratified sampling: If the population from which a sample is to be drawn does not constitute a homogeneous group, then stratified sampling technique is applied so as to obtain a representative sample.

(v) Quota sampling:In stratified sampling the cost of taking random samples from individual strata is often so expensive that interviewers are simply given quota to be filled from different strata, the actual selection of items for sample being left to the interviewer's judgment.

(vii) Multi-stage sampling:
This is a further development of the idea of cluster sampling.(viii) Sequential sampling:This is somewhat a complex sample design where the ultimate size of the sample is not fixed in advance but is determined according to mathematical decisions on the basis of information yielded as survey progresses.

6. Collecting the data:Data can be collected by any one or more of the following ways:(i) By observation(ii) Through personal interview(iii) Through telephone interviews(iv) By mailing of questionnaires(v) Through schedules

7. Execution of the project:
8. Analysis of data:Requires a number of closely related operations such as establishment ofCategoriesCodingEditingTabulationAnalysis work after tabulation is generally based on the computation of various percentages, coefficients, etc., by applying various well defined statistical formulae.

9. Hypothesis-testing:Various tests, such as Chi square test, t-test, F-test, have been developed by statisticians for the purpose.Hypothesis-testing will result in either accepting the hypothesis or in rejecting it.

10. Generalisations and interpretation 11

10. Generalisations and interpretation Preparation of the report or the thesis

3 Research Designoverall research design into the following parts:(a) the sampling design .(b) the observational design .(c) the statistical design .(d) the operational design which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.

NEED FOR RESEARCH DESIGN

Research design is needed because it facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money.

FEATURES OF A GOOD DESIGN

A research design appropriate for a particular research problem,usually involves the consideration of the following factors:(i) the means of obtaining information;(ii) the availability and skills of the researcher and his staff, if any.(iii) the objective of the problem to be studied.(iv) the nature of the problem to be studied.(v) the availability of time and money for the research work.

IMPORTANT CONCEPTS RELATING TO RESEARCH DESIGN

1. Dependent and independent variables:A concept which can take on different Quantitative values is called a variable. As such the concepts like weight, height, income are all examples of variables. Qualitative phenomena (or the attributes) are also quantified on the basis of the presence or absence of the concerning attribute(s).'continuous variables'. Age *'discrete variables'. sex**For instance, if we say that height depends upon age, then height is a dependent variable and age is an independent variable.

2. Extraneous variable: Independent variables 3

2.Extraneous variable: Independent variables 3.Control: One important characteristic of a good research design is to minimize the influence or effect of extraneous variable(s). 4.Confounded relationship: When the dependent variable is not free from the influence of extraneous variable(s), the relationship between the dependent and independent variables is said to be confounded by an extraneous variable(s). 5.Research hypothesis: When a prediction or a hypothesized relationship is to be tested by scientific methods, it is termed as research hypothesis. The research hypothesis is a predictive statement that relates an independent variable to a dependent variable.

6.Experimental and non-experimental hypothesis-testing research: When the purpose of research is to test a research hypothesis, it is termed as hypothesis-testing research. It can be of the experimental design or of the non-experimental design.

7. Experimental and control groups: In an experimental hypothesis-testing research when a group is exposed to usual conditions, it is termed a 'control group', but when the group is exposed to some novel or special condition, it is termed an 'experimental group'.

8. Treatments:The different conditions under which experimental and control groups are put are usually referred to as 'treatments'.9. Experiment:The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.

10. Experimental unit(s): The pre-determined plots or the blocks, where different treatments are used, are known as experimental units. Such experimental units must be selected (defined) very carefully.

DIFFERENT RESEARCH DESIGNS

Different research designs can be conveniently described if we categorize them as:(1) Exploratory research studies; Exploratory research studies are also termed as formulative research studies.(2) Descriptive And Diagnostic research studies, (((survey design ; Several methods (viz., observation, questionnaires, interviewing, examination of records, Cohort {incidence},cross sectional {prevalence}, and case-control studies etc.),Descriptive research studies are those studies which are concerned with describing the characteristics of a particular individual, or of a group,whereas diagnostic research studies determine the frequency with which something occurs or its association with something else.(3) Hypothesis-testing research studies. Hypothesis-testing research studies (generally known as experimental studies) are those where the researcher tests the hypotheses of causal relationships between variables. Such studies require procedures that will not only reduce bias and increase reliability, but will permit ; agricultural research (such as treatment, yield, plot, block etc.)

BASIC PRINCIPLES OF EXPERIMENTAL DESIGNS :

Professor Fisher has enumerated three principles of experimental designs:(1) the Principle of Replication;(2) the Principle of Randomization;(3) Principle of Local Control.

Important Experimental Designs

(a) Informal experimental designs:(i) Before-and-after without control design. single test group or area(ii) After-only with control design. two groups or areas (test area and

control area)(iii) Before-and-after with control design. two areas are selected and the dependentvariable is measured in both the areas for an identical time-period before the treatment.

(ii) After-only with control design.

(b) Formal experimental designs:

(i) Completely randomized design (C.R. Design).Simple design. ; Involves only two principles viz., the principle of replication and the principle of randomization of experimental designs. (one-way ANOVA)*(1) Two-group simple randomized design:Population Randomly selected, Sample Randomly assigned Experimental group Treatment A & Control group reatment B (both are Independent variable)(2) Random replications design: The limitation of the two-group randomized design is usuallyeliminated within the random replications design.

(ii) Randomized block design (R. B. Design). In the R. B

(ii) Randomized block design (R.B. Design) In the R.B. design the principle of local control can be applied along with the other two principles of experimental designs. In the R.B. design, subjects are first divided into groups, known as blocks, such that within each group the subjects are relatively homogeneous in respect to some selected variable. (two-way ANOVA)*

(iii) Latin square design (L. S. Design)

(iii) Latin square design (L.S. Design). is an experimental design very frequently used in agricultural research. (two-way ANOVA)*(iv) Factorial designs.(a) Simple factorial designs:Simple factorial design may either be a 2 × 2 simple factorial design, or it may be, say, 3 × 4 or 5 × 3 or the like type of simple factorial design.

(iii) l.s. design

(iv) Factorial designs Simple Factorial designs

The data obtained in case of two (2 × 2) simple factorial studies may be as given in Fig. 3.9.

2 × 2 simple factorial design

(4 × 3 simple factorial design).

(b) Complex factorial designs:

Experiments with more than two factors at a time involve the use of complex factorial

designs.the design used will be termed 2 × 2 × 2 complex factorial design which will contain a total of eight cell

Complex factorial designs:

CONCLUSIONThere are several research designs and the researcher must decide in advance of collection andanalysis of data as to which design would prove to be more appropriate for his research project.

4 Sampling Design CENSUS AND SAMPLE SURVEY STEPS IN SAMPLE DESIGN

(i) Type of universe:(ii) Sampling unit:.Sampling unit may be a geographical one such as state, district, village, etc., or a construction unit such as house, flat, etc., or it may be a social unit such as family, club, school, etc.(iii) Source list:It is also known as 'sampling frame' from which sample is to be drawn. It contains the names of all items of a universe (in case of finite universe only).(iv) Size of sample:This refers to the number of items to be selected from the universe to constitute a sample.(v) Parameters of interest:the specific population parameters which are of interest. For instance, we may be interested in estimating the proportion of persons with some characteristic in the .(vi) Budgetary constraint: Cost considerations, from practical point of view, have a major impact upon decisions relating to not only the size of the sample but also to the type of sample.(vii) Sampling procedure: Finally, the researcher must decide the type of sample he will use i.e., he must decide about the technique to be used in selecting the items for the sample.

Sampling errors are the random variations in the sample estimates around the true population parameters.Since they occur randomly and are equally likely to be in either direction, their nature happens to be of compensatory type and the expected value of such errors happens to be equal to zero.Sampling error decreases with the increase in the size of the sample, and it happens to be of a smaller magnitude in case of homogeneous population.

DIFFERENT TYPES OF SAMPLE DESIGNS

Sample designs are basically of two types viz., non-probability sampling and probability sampling.Non-probability sampling : deliberate sampling, purposive sampling and judgment sampling; 'non-random'; Quota sampling.Probability sampling : 'random sampling' or 'chance sampling'. Under this sampling design, every item of the universe has an equal chance of inclusion in the sample. Lottery method

HOW TO SELECT A RANDOM SAMPLE ?

Simple cases: slip of paper; random number tables; diceCOMPLEX RANDOM

SAMPLING DESIGNSSystematic sampling:Stratified samplingCluster sampling: :Area sampling:Multi-stage sampling:Sampling with probability proportional to size:Sequential sampling:

5 Measurement and Scaling Techniques

MEASUREMENT IN RESEARCH,In our daily life we are said to measure when we use some yardstick to determine weight, height, or some other feature of a physical object.

MEASUREMENT SCALESFrom what has been stated above, we can write that scales of measurement can be considered in terms of their mathematical properties.The most widely used classification of measurement scales are(a) nominal scale; system of assigning number symbols to events in order to label them. ex-post-facto research(b) ordinalscale; The lowest level of the ordered scale that is commonly used is the ordinal scale. qualitative phenomena.

(c) Interval scale: The Fahrenheit scale is an example

(c) Interval scale: The Fahrenheit scale is an example. (d) Ratio scale: Ratio scale represents the actual amounts of variables. Measures of physical dimensions such as weight, height, distance, etc. are examples.

Scale Classification Bases

(a) Subject orientation: to measure characteristics of the respondent. (b) Response form: classify the scales as categorical and comparative.Categorical scales (c) Degree of subjectivity: With this basis the scale data may be based on whether we measure subjective personal preferences or simply make non-preference (d) Scale properties: Considering scale properties, one may classify the scales as nominal, ordinal, interval and ratio scales. (e) Number of dimensions: In respect of this basis, scales can be classified as ‘one-dimensional’ and ‘multidimensional’ scales.

(f) Scale construction techniques: Following are the five main techniques by which scales can be developed. (i) Arbitrary approach: It is an approach where scale is developed on ad hoc basis. This is the most widely used approach. (ii) Consensus approach: Here a panel of judges evaluate the items chosen for inclusion in the instrument in terms of whether they are relevant to the topic area and unambiguous in implication. (iii) Item analysis approach: Under it a number of individual items are developed into a test which is given to a group of respondents. After administering the test, the total scores are calculated for every one. Individual items are then analyzed to determine which items discriminate between persons or objects with high total scores and those with low scores. (iv) Cumulative scales are chosen on the basis of their conforming to some ranking of items with ascending and descending discriminating

power. For instance, in such a scale the endorsement of an item representing an extreme position should also result in the endorsement of all items indicating a less extreme position. (v) Factor scales may be constructed on the basis of inter correlations of items which indicate that a common factor accounts for the relationship between items. This relationship is typically measured through factor analysis method.

6 Methods of Data Collection

The methods of collecting primary and secondary data differ since primary data are to be originally collected, while in case of secondary data the nature of data collection work is merely that of compilation.

COLLECTION OF PRIMARY DATA

Important ones are: (i) observation method, (ii) interview method, (iii) questionnaires, (iv) schedules, (v) other methods which include (a) warranty cards; (b) distributor audits; (c) pantry audits; (d) consumer panels; (e) using mechanical devices; (f) through projective techniques; (g) depth interviews, and (h) content analysis

Observation MethodThe observation method is the most commonly used method specially in studies relating to behavioral sciences.Observation becomes a scientific tool and the method of data collection for the researcher, when it serves a formulated research purpose, is systematically planned and recorded and is subjected to checks and controls on validity and reliability.The main advantage of this method is that subjective bias is eliminated, if observation is done accurately.structured observation.unstructured observation.participant observation.non-participant observation. (disguised observation.)

Interview MethodThe interview method of collecting data involves presentation of oral-verbal stimuli and reply in terms of oral-verbal responses.This method can be used through personal interviews and, if possible, through telephone interviews.

DIFFERENCE BETWEEN QUESTIONNAIRES AND SCHEDULES

Both questionnaire and schedule are popularly used methods of collecting data in research surveys.There is much resemblance in the nature of these two methods and this fact has made many peopleto remark that from a practical point of view, the two methods can be taken to be the same But from the technical point of view there is difference between the two.

SOME OTHER METHODS OF DATA COLLECTION

1. Warranty cards:2. Distributor or store audits:3. Pantry audits:4. Consumer panels:5. Use of mechanical devices:6. Projective techniques:7. Depth interviews:8. Content-analysis